Supplementary Material

Record-Level Evaluation of a Morlet CNN–Conformer for Five-Class ECG Beat Classification

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1 Extended Methods

1.1 Record partition and preprocessing

The analysis included 45 MIT–BIH records, omitting records 102, 104, and 107. A seeded record-level split assigned 38 records to development and seven (106, 207, 208, 220, 228, 231, and 233) to testing. Record 217 remained in the development set. The five cross-validation partitions were generated by shuffling the 38 development records with the same seed and dividing them into five folds. Hyperparameter search used the first of these partitions as its training–validation split; the test records remained untouched until final model evaluation.

Beat locations and symbols were read from the reference annotation files. For accepted symbols, 187 samples were taken from the first signal channel, centred on the annotation. The continuous wavelet transform used the Morlet wavelet and integer scales 1 through 32; coefficient magnitudes, rather than signed or complex coefficients, were retained. Three consecutive scalograms within a record formed each input, and the third beat supplied the class label. The first two beat sequences that could not be completed at the beginning of each record were therefore absent from evaluation.

Normalisation was estimated separately for every training partition. Scalograms were reshaped so that the 187 within-beat sample positions were standardised across beats and wavelet scales. Training means and standard deviations were then applied unchanged to validation data. For final fitting, the estimates came from all 38 development records and were applied to the seven test records.

1.2 Architecture and optimisation details

The shared beat encoder contained three 3×3 convolutions. Filter counts were 32, 64, and the selected embedding dimension; all convolutions were followed by batch normalisation and ReLU activation, and the first two were followed by 2×2 max pooling. Global average pooling produced one vector per beat.

The selected CNN–Conformer used a 128-dimensional embedding, eight attention heads, a feed-forward expansion factor of four, kernel width 11, dropout 0.2, and one Conformer block. The selected CNN–attention model used a 64-dimensional embedding, eight heads, a feed-forward expansion factor of four, two attention/feed-forward blocks, and no dropout. The CNN–LSTM used a 128-dimensional embedding and 128 recurrent units. The CNN-only model used a 128-dimensional embedding and averaged the three beat vectors.

Hyperband trials ran for at most 20 epochs. Early stopping monitored validation loss with patience five; the learning rate was reduced by a factor of 0.2 after one epoch without improvement, to a minimum of 10^{-6}. Final fits used AdamW at 10^{-5}, sparse categorical cross-entropy, balanced class weights, batches of 128, and 20 epochs. No early stopping was applied to the final fits.

2 Supplementary Analyses

2.1 Cross-validation

Accuracy for the selected CNN–Conformer varied from 0.210 to 0.635 across the five record-level development folds (Table 1). The fold results establish substantial sensitivity to which records were used for validation. Class-specific predictions were not retained for every fold, so the source of this variation cannot be assigned to a particular rhythm class.

Table 1: Five-fold cross-validation performance for the CNN–Conformer. Metrics correspond to the record-level development folds used during model assessment.

Fold	Accuracy	Loss
1	0.539	0.949
2	0.210	1.656
3	0.529	1.384
4	0.294	1.738
5	0.635	0.839
Mean ± SD	0.442 ± 0.161	1.313 ± 0.363

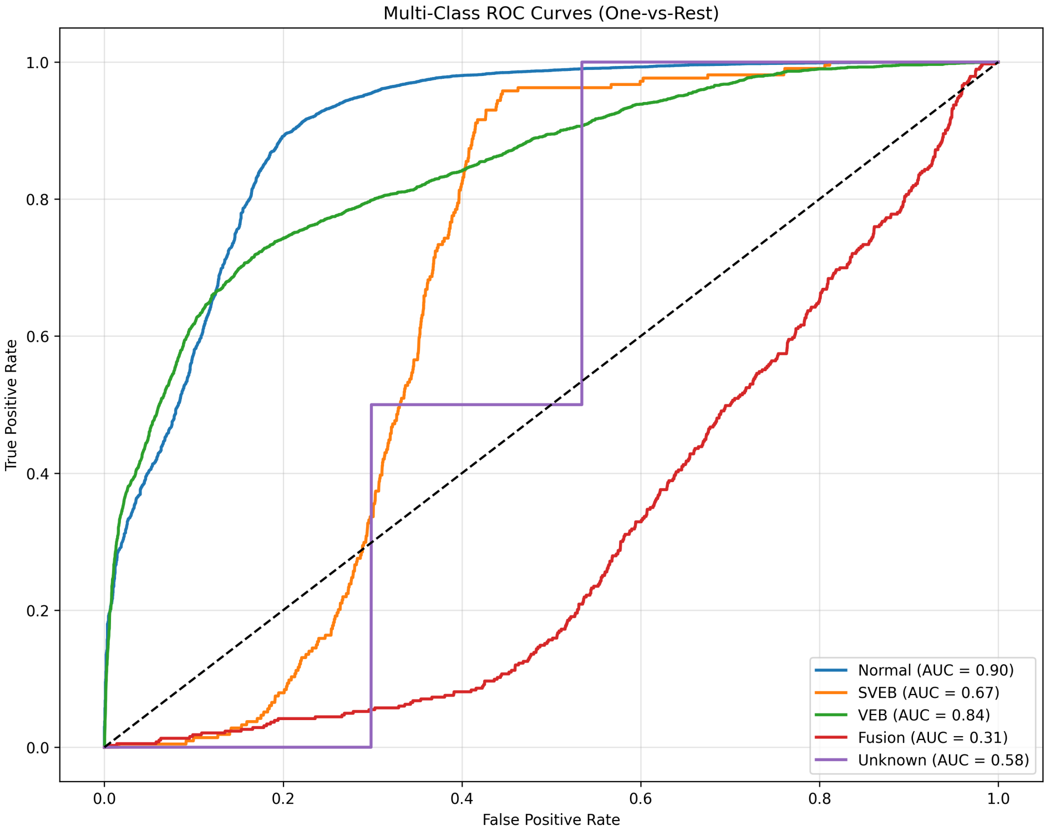
2.2 Threshold analysis

One-vs-rest thresholds were examined after final prediction. The largest observed F1 was 0.06 for SVEB (threshold 0.09; precision 0.03; recall 0.90) and 0.05 for Fusion (threshold approaching zero; precision 0.02; recall 1.00). These operating points were selected and evaluated on the same seven records. They describe the score distributions in this test set and should not be used as prospective thresholds.

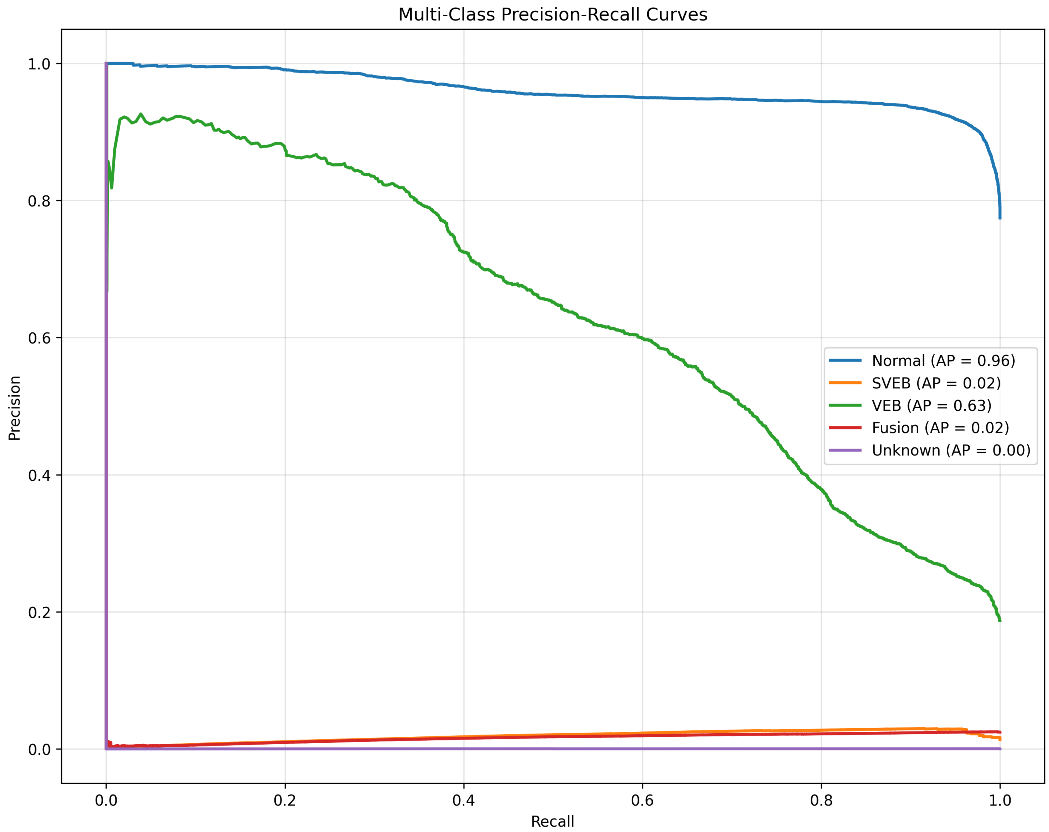
2.3 Calibration diagnostic

Before temperature scaling, 15-bin expected calibration error was 0.417 for Normal, 0.098 for SVEB, 0.089 for VEB, 0.103 for Fusion, and 0.153 for Q. The corresponding Brier scores were 0.289, 0.039, 0.121, 0.049, and 0.051. A global temperature of 1.050 changed mean negative log-likelihood from 1.047989 to 1.047731. The temperature was fitted on the test set; these values are therefore an internal diagnostic rather than an independent calibration assessment.

3 Supplementary Figures and Tables



Supplementary Figure S1: One-vs-rest receiver operating characteristic curves for the CNN–Conformer on the seven test records. The Q curve is shown only as a descriptive consequence of the two available Q targets.



Supplementary Figure S2: One-vs-rest precision–recall curves for the CNN–Conformer on the seven test records.